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Behavioral Finance in Cryptocurrency Perpetual Futures and Swaps: A Systematic Literature Review

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Abstract

This review synthesizes the emerging literature on behavioral finance in cryptocurrency perpetual futures and perpetual swaps. It uses a constrained systematic review of accessible repositories, publisher pages, and citation trails for studies published or posted from January 2021 to April 2026. The synthesis separates 13 direct perpetual-futures studies from 6 adjacent behavioral studies that inform interpretation. The central question is how behavioral mechanisms shape trading, pricing, and market quality in perpetual futures markets. The strongest direct evidence concerns speculative demand and basis risk, leverage choice and liquidation risk, funding-rate carry and arbitrage behavior, informed trading and market quality, and exchange-design effects across centralized and decentralized venues. Direct evidence on classic behavioral constructs such as fear of missing out, overconfidence, disposition effects, and learning remains sparse in perpetual-specific settings. Three conclusions stand out. First, perpetuals are behaviorally distinctive because funding fees, leverage, mark-to-market margining, and auto-liquidation create a high-frequency feedback system between prices and trader positions. Second, the strongest causal evidence indicates that perpetual contracts increase spot-market trading volume but worsen adverse-selection conditions when informed trading rises around funding windows. Third, exchange design matters because inverse, linear, quanto, oracle-priced, and VAMM-based contracts expose traders to different incentives and liquidation dynamics. The most important research gaps concern trader-level identification, CEX–DEX comparisons using comparable data, contract-type heterogeneity, and causal tests of leverage-rule changes.

Keywords: perpetual futures; perpetual swaps; cryptocurrency; behavioral finance; funding rates; liquidation; leverage; arbitrage; informed trading; market quality.

JEL codes: G13, G14, G41, G12, D91

Introduction and Motivation

Perpetual futures, also called perpetual swaps, are derivative contracts without fixed maturity that use recurring funding transfers to keep contract prices near spot prices (Ackerer et al., 2025; Neubert et al., 2026a; Shiller, 1993). The contract design differs from dated futures because no terminal delivery date mechanically forces convergence to spot (Ackerer et al., 2025; He et al., 2022; Neubert et al., 2026b). Academic pricing research describes perpetuals as the most important cryptocurrency derivative because they provide leveraged exposure without rollover and because funding payments create a continuous anchoring mechanism (He et al., 2022; Gornall et al., 2024).

Perpetual contracts create a distinct behavioral-finance setting because beliefs, leverage, funding, liquidations, and arbitrage constraints interact before any terminal settlement can discipline prices (He et al., 2022; Gornall et al., 2024). Funding rates can reveal directional demand because long and short positions periodically transfer cash when the contract trades away from spot (Christin et al., 2023; Schmeling et al., 2023; He et al., 2022). Liquidation rules can trigger forced position closures in response to small price movements when traders use high leverage and thin collateral buffers (Cheng et al., 2021; Alexander et al., 2023).

The behavioral question is therefore narrower than the general question of whether cryptocurrency markets respond to sentiment (Almeida & Gonçalves, 2023). The central issue is how perpetual-contract architecture translates sentiment, disagreement, leverage demand, and arbitrage limits into observable outcomes such as funding-rate dynamics, basis deviations, liquidations, open interest, informed trading, market quality, and spot–perpetual spillovers (Cheng et al., 2021; Ruan & Streltsov, 2022; He et al., 2022; Chen et al., 2024). This review keeps perpetual-specific evidence at the center. It uses broader crypto-behavior studies only

when they sharpen the interpretation of perpetual-linked trading, derivatives positioning, leverage, liquidation, funding-rate arbitrage, or spot–perpetual interaction (Almeida & Gonçalves, 2023; Kogan et al., 2023; Dunbar & Owusu-Amoako, 2023).

The contribution is threefold. First, the review isolates perpetual-contract architecture as a behavioral transmission mechanism rather than treating cryptocurrency investor behavior as a single market-level phenomenon (Almeida & Gonçalves, 2023; He et al., 2022). Second, the review maps behavioral-finance constructs to the market-level proxies that perpetual-futures papers actually observe, including basis, funding, open interest, liquidations, spreads, VPIN, and carry returns (Easley et al., 2012; Cheng et al., 2021; Ruan & Streltsov, 2022; He et al., 2022). Third, the review distinguishes direct, perpetual evidence from adjacent behavioral evidence to ensure that psychological interpretations do not exceed the data that support them (Almeida & Gonçalves, 2023; Chen et al., 2024; Gornall et al., 2024).

Research Questions

The review uses four research questions to connect screening, coding, synthesis, and interpretation. This structure follows review-reporting logic that requires explicit eligibility criteria and transparent synthesis categories (Neubert, 2026; Page et al., 2021).

1. RQ1. Which behavioral mechanisms does the direct perpetual-futures literature identify in trading, pricing, funding, liquidation, and market quality?
2. RQ2. How do contract design and venue architecture transmit trader behavior into funding rates, basis deviations, liquidations, and market-quality outcomes?
3. RQ3. Which methods provide the strongest evidence, and how does the evidential strength differ between direct perpetual studies and adjacent cryptocurrency-behavior studies?

4. RQ4. Which research gaps remain for trader-level identification, CEX–DEX comparison, contract-type heterogeneity, leverage-rule changes, and regime dependence?

Theoretical Framework

The review draws on behavioral asset-pricing theory, prospect theory, overconfidence models, and limits-to-arbitrage theory because perpetual contracts combine risky choice, leveraged trading, and constrained arbitrage within a single market structure (Kahneman & Tversky, 1979; Odean, 1998; Shleifer & Vishny, 1997; He et al., 2022). Prospect theory supports the interpretation of liquidation-aware behavior because traders may weight losses asymmetrically when margin losses move positions toward forced closure (Kahneman & Tversky, 1979; Alexander et al., 2023). Loss-aversion interpretations remain only semi-direct in this literature because most perpetual studies observe liquidation outcomes and margin mechanics rather than trader-specific utility functions (Alexander et al., 2023; Cheng et al., 2021).

Overconfidence theory supports the interpretation of excessive turnover, aggressive leverage, and poor calibration between risk and expected performance (Odean, 1998; Barber & Odean, 2000). In perpetual markets, high leverage and repeated liquidation exposure provide indirect evidence that some traders accept extreme downside risk in exchange for lottery-like upside exposure (Alexander et al., 2023; Cheng et al., 2021). This interpretation remains cautious because high leverage can also reflect hedging conventions, collateral constraints, or short-horizon arbitrage rather than psychological overconfidence alone (Alexander et al., 2023; He et al., 2022).

Noise-trader and limits-to-arbitrage theory support the interpretation of persistent basis deviations and carry returns in perpetual markets (De Long et al., 1990; Shleifer & Vishny,

1997). The perpetual-futures literature shows that basis deviations can remain economically meaningful when speculative demand, capital constraints, and market segmentation prevent arbitrageurs from fully closing gaps between spot and derivative prices (Christin et al., 2023; Schmeling et al., 2023; He et al., 2022; Gornall et al., 2024). This framework also explains why funding rates can reduce basis risk without eliminating it when arbitrage capital becomes scarce during stress (Gornall et al., 2024; Joshi & Chance, 2026).

Attention, sentiment, and disagreement theories enter the review through revealed behavior rather than survey measurement because most perpetual datasets contain prices, funding, open interest, volume, liquidation, and order-book variables (Almeida & Gonçalves, 2023; Giagkiozis & Said, 2024; Ruan & Streltsov, 2022). Broader crypto studies show that retail crypto traders respond to momentum, speculative positioning, and asset-specific attention, which makes those mechanisms relevant but indirect for perpetual markets (Dunbar & Owusu-Amoako, 2023; Kogan et al., 2023; Liu et al., 2022). The review, therefore, treats FOMO, overconfidence, herding, and disagreement as interpretive constructs unless the focal study observes trader identities, positions, or direct behavioral measures (Almeida & Gonçalves, 2023; Chen et al., 2024).

Review Protocol and Methodology

The study uses a systematic literature review following the PRISMA protocol (Neubert, 2022a, 2022b). It is a qualitative research method (Neubert, 2025). The search relies on accessible publisher pages, repositories, working paper servers, conference pages, and citation trails rather than on native Web of Science and Scopus (Page et al., 2021; Table 1). This designation avoids overstating database completeness while retaining transparent screening, coding, and source-verification procedures (Page et al., 2021; Appendix A).

Searches were executed on 6–7 April 2026 with three string families: (“perpetual future*” OR “perpetual swap*” OR perp*) AND (crypto* OR bitcoin OR ether* OR digital asset*); (“perpetual future*” OR “perpetual swap*”) AND (behavior* OR sentiment OR attention OR herding OR disagreement OR momentum OR overconfidence OR liquidation OR leverage OR funding rate OR basis OR arbitrage OR informed trading OR market quality); and (bitcoin futures OR crypto derivatives) AND (retail trading OR investor attention OR sentiment OR speculative demand OR trader behavior) (Table 1; Appendix A). The accessible sources were SSRN, NBER, BIS, publisher pages from Elsevier, Wiley, Springer, Taylor & Francis, Ledger, AEA, or author pages, and backward or forward citation trails (Appendix A).

The review included a study in the direct perpetual set when it addressed perpetual futures or perpetual swaps directly and contained evidence on pricing, funding, leverage, liquidation, arbitrage, contract design, market quality, trader behavior, or stress dynamics (Table 1). The review retained adjacent behavioral studies only when they supplied direct evidence about cryptocurrency investor behavior, futures positioning, trader heterogeneity, learning, or risk factors that helped interpret perpetual-futures mechanisms (Table 2). Purely legal, regulatory, or mechanical pricing sources were excluded from the focal behavioral synthesis unless they clarified the institutional design of perpetual contracts (Table 1; Appendix A).

The corrected search accounting separates direct perpetual studies from adjacent behavioral studies to remove the earlier sample-count inconsistency (Table 1). The search identified 47 records, removed 9 duplicates or overlapping versions, screened 38 titles and abstracts, assessed 20 full texts for direct or near-direct relevance, excluded 7 full texts, and retained 13 direct perpetual studies (Table 1). The review then retained 6 adjacent behavioral

studies for interpretive triangulation, producing a total synthesis set of 19 studies (Table 1; Table 2).

Each included study was coded on five dimensions: identification strength, data quality, market coverage, behavioral-construct clarity, and external validity (Tables 2 and 4). The strongest designs exploit institutional timing, funding windows, contract introductions, exchange terminations, high-frequency microstructure data, or quasi-experimental designs (Ruan & Streltsov, 2022; Alexander et al., 2024; Giagkiozis & Said, 2024). Structural and no-arbitrage papers provide a benchmark for discipline but require empirical pairing to identify psychological behavior directly (He et al., 2022; Ackerer et al., 2025; Gornall et al., 2024).

Table 1

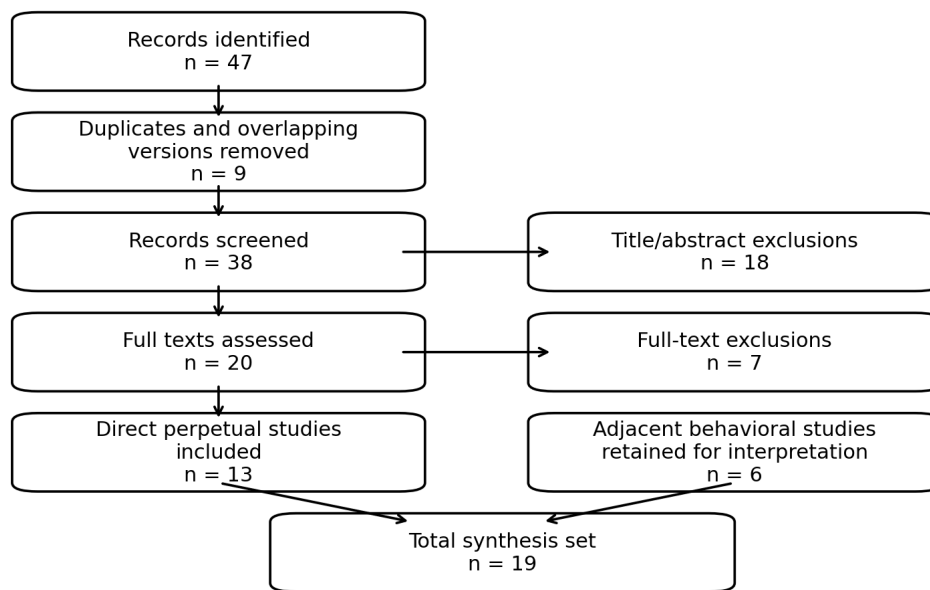
Search Strategy and Screening Summary

Component	Details
Search and screening principle	All identified records were screened with the same eligibility rules and then classified as direct, adjacent, background, or excluded.
Search dates	6–7 April 2026.
Accessible sources searched	SSRN, NBER, BIS, publisher pages, Ledger, AEA, or author pages, and backward/forward citation trails.
Records identified	47 records before de-duplication.
Duplicates/overlapping versions removed	9 records.
Records screened	38 title/abstract screens.
Title/abstract exclusions	18 records.
Full texts assessed for direct or near-direct relevance	20 full texts.
Full-text exclusions	7 studies: legal/regulatory only, pricing/mechanics only, or too indirect for perpetual-specific behavioral inference.
Direct perpetual studies included	13 studies.
Adjacent behavioral studies retained for interpretation	6 studies.
Total synthesis set	19 studies.
Verification rule	Each retained source was checked against a DOI, publisher page, SSRN record, NBER page, BIS page, conference page, or author page where available.

Note. The flow follows PRISMA-style logic but should be read as a constrained systematic review because Web of Science and Scopus were not queried natively (Page et al., 2021).

Figure 1

Search and Screening Flow



Note. The figure reconciles the final sample by separating 13 direct perpetual studies from 6 adjacent behavioral studies, which were retained for interpretive triangulation (Table 1).

Table 2

Concise Evidence Classification

Study	Status and setting	Main channel	Role in review
Cheng et al. (2021)	Applied Economics; BitMEX BTC perpetuals	Leverage, liquidation, tail risk, optimal margin	Direct empirical
Alexander et al. (2023)	European Journal of Operational Research; Bitcoin futures	Liquidation-aware hedging and leverage selection	Direct empirical/theoretical
Christin et al. (2023)	Working paper; crypto carry trade	Funding-rate carry, leverage demand, and disagreement	Direct empirical
Schmeling et al. (2023)	BIS Working Paper; crypto futures/perps	Carry, segmentation, arbitrage limits	Direct empirical
Giagkiozis & Said (2024)	Ledger; BTC perpetual swaps across exchanges	Open-interest quality, liquidation reporting, exchange data reliability	Direct empirical
Chen et al. (2024)	Electronic Markets; CEX and DEX perpetuals	Exchange design, long/short asymmetry, VAMM/oracle effects	Direct empirical/taxonomic
Alexander et al. (2024)	Journal of Financial Markets, Options, and Perpetuals	Efficiency, arbitrage, trader composition	Direct empirical
He et al. (2022)	SSRN/arXiv; major crypto perpetuals	No-arbitrage benchmarks, deviations, and carry returns	Direct theory + empirics
Ruan & Streltsov (2022)	SSRN working paper; spot/perpetual markets	Funding-cycle behavior, informed trading, market quality	Direct causal
Ackerer et al. (2025)	Mathematical Finance; perpetual pricing	Funding design, linear/inverse/quanto structure	Direct theory/design
Gornall et al. (2024)	SSRN/AFA working paper; futures design	Speculative demand, basis risk, crisis liquidity	Direct theory + empirics
Zhang (2026)	SSRN working paper; funding-rate mechanism	Funding as an algorithmic feedback rule	Direct theory

Study	Status and setting	Main channel	Role in review
Joshi & Chance (2026)	SSRN working paper; stress and runs	Adverse selection, stress thresholds, open-interest runs	Direct theory/empirical design
Almeida & Gonçalves (2023)	Journal of Behavioral and Experimental Finance; SLR	Sentiment, herding, investor behavior	Adjacent behavioral
Dunbar & Owusu-Amoako (2023)	Journal of Behavioral and Experimental Finance; futures positioning	Speculative retail futures behavior and return predictability	Adjacent behavioral
Ferko et al. (2023)	Global Finance Journal; CME BTC futures	Trader heterogeneity and participant types	Adjacent behavioral
Gemayel & Preda (2021)	International Review of Financial Analysis; trader behavior	Learning, ambiguity, gambler's fallacy	Adjacent behavioral
Kogan et al. (2023)	NBER Working Paper; retail accounts	Retail momentum, adoption beliefs, crypto-stock differences	Adjacent behavioral
Liu et al. (2022)	Journal of Finance; crypto asset pricing	Common risk factors and momentum background	Adjacent asset-pricing

Note. The table separates direct perpetual studies from adjacent studies to avoid treating indirect behavioral evidence as direct perpetual-futures evidence.

Conceptual Foundations: Why Perpetual Futures Are Behaviorally Distinctive

Perpetual futures differ from standard dated futures in four behaviorally important ways: they lack a maturity date, use recurring funding transfers, permit high leverage, and rely on margin and liquidation rules that operate continuously (Neubert et al., 2026a, 2026b). These features make the contract a high-frequency laboratory for belief disagreement, leverage demand, arbitrage constraints, and forced liquidation (Christin et al., 2023; Ruan & Streltsov, 2022; Gornall et al., 2024). The design matters because the same contract can serve directional speculation, hedging, market making, and funding-rate carry, which means observed market variables aggregate heterogeneous motives (Ferko et al., 2023; Alexander et al., 2024; Chen et al., 2024).

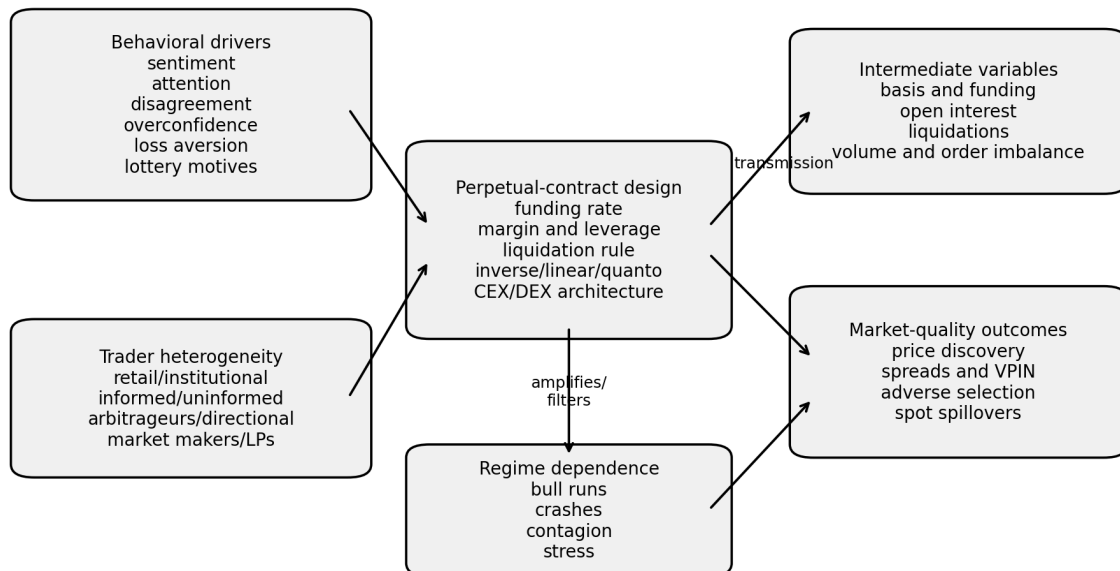
Funding rates translate long-short imbalances into recurring cash flows, so funding becomes both a price-alignment mechanism and a proxy for revealed demand (Christin et al., 2023; He et al., 2022; Akerer et al., 2025). Leverage transforms small spot-price movements into forced exits when collateral falls below maintenance thresholds, thereby amplifying price pressure during periods of volatility (Cheng et al., 2021; Alexander et al., 2023). Exchange architecture changes the transmission mechanism because CEX order books, oracle-priced

DEXs, and VAMM designs differ in how they form prices, provide liquidity, and define trader roles (Chen et al., 2024).

Behavior in the perpetual-futures literature is usually observed through market microstructure rather than direct psychology (Almeida & Gonçalves, 2023; Giagkiozis & Said, 2024; Ruan & Streltsov, 2022). Researchers typically proxy sentiment, attention, disagreement, overconfidence, and informed trading with funding rates, basis, open interest, liquidation frequency, order imbalance, spreads, and VPIN (Easley et al., 2012; Cheng et al., 2021; He et al., 2022; Ruan & Streltsov, 2022). This proxy structure requires cautious inference because the same observable variable can reflect sentiment, arbitrage constraints, inventory pressure, or data-reporting quality (Giagkiozis & Said, 2024; He et al., 2022).

Figure 2

Behavioral Channels in Cryptocurrency Perpetual Futures Markets



Note. The map synthesizes direct perpetual-futures studies and adjacent behavioral evidence; the arrows represent interpretation rather than causal proof (Cheng et al., 2021; Chen et al., 2024; He et al., 2022; Ruan & Streltsov, 2022; Gornall et al., 2024).

Thematic Literature Review

Each thematic subsection begins with direct perpetual evidence and then uses adjacent cryptocurrency-behavior evidence only when it clarifies interpretation (Table 2). This hierarchy matters because market-level proxies cannot by themselves identify individual psychological motives without trader-level or account-level data (Almeida & Gonçalves, 2023; Chen et al., 2024; He et al., 2022).

Sentiment, Attention, and Disagreement

Direct, perpetual evidence of sentiment remains thin because most studies observe funding, basis, open interest, spreads, and order flow rather than surveying sentiment or traders' beliefs (He et al., 2022; Ruan & Streltsov, 2022; Giagkiozis & Said, 2024). Ruan and Streltsov (2022) show that spot-market volume and quoted spreads rise around the eight-hour funding cycle, which implies that funding windows attract trading activity and adverse selection. He et al. (2022) show that perpetual futures can deviate materially from no-arbitrage benchmarks, comove with one another across cryptocurrencies, and remain economically meaningful despite declining over time.

These findings support a revealed-demand interpretation rather than a direct psychological interpretation (He et al., 2022; Ruan & Streltsov, 2022). Basis and funding movements can reflect common sentiment, common funding conditions, common speculative demand, market-maker inventory pressure, or constrained arbitrage capital (Christin et al., 2023; Schmeling et al., 2023; He et al., 2022). For that reason, the review treats funding rates and the

basis as proxies for indirect sentiment and disagreement unless a study combines them with trader-level data or stronger identification (He et al., 2022; Ruan & Streltsov, 2022).

Adjacent evidence strengthens the interpretation but does not replace direct perpetual evidence (Table 2). Kogan et al. (2023) show that retail crypto traders behave more like momentum traders in cryptocurrencies than in stocks or gold. Dunbar and Owusu-Amoako (2023) show that speculative retail futures positioning predicts cryptocurrency returns after controlling for attention, sentiment, and uncertainty. These results imply that perpetual markets may inherit a retail-attention and momentum component from the broader cryptocurrency ecosystem, but the inference remains indirect when the focal data do not observe perpetual accounts directly (Kogan et al., 2023; Dunbar & Owusu-Amoako, 2023).

Herding, Momentum, and Feedback Trading

Broader research on crypto behavior documents momentum-like retail trading, social-sentiment effects, and herding-like dynamics in cryptocurrency markets (Almeida & Gonçalves, 2023; Kogan et al., 2023; Liu et al., 2022). These mechanisms matter for perpetuals because funding rates, open interest, and liquidations provide high-frequency channels through which feedback trading can appear in market data (He et al., 2022; Gornall et al., 2024). The direct perpetual evidence remains more convincing when it links feedback to basis, funding, or contract-design features rather than to generic crypto price movements (Christin et al., 2023; Gornall et al., 2024).

Gornall et al. (2024) model a setting in which volatile speculative demand interacts with constrained arbitrage capital, pushing futures prices away from spot prices. The same paper argues that frequent funding payments reduce extreme basis risk and improve liquidity relative to dated futures when speculative demand becomes volatile (Gornall et al., 2024). This result links

behavioral demand to contract design because funding payments do not eliminate speculative demand; rather, they change how that demand manifests in basis risk and liquidity (Gornall et al., 2024; Ackerer et al., 2025).

Frontier theory papers extend the feedback mechanism but currently provide weaker behavioral identification than mature empirical papers (Zhang, 2026; Joshi & Chance, 2026). Zhang (2026) models funding as an algorithmic feedback rule that can dampen or amplify basis volatility depending on design. Joshi and Chance (2026) model stress-driven runs in open interest when adverse-selection costs interact with public and private signals. These working papers support future theoretical development, but their conclusions should remain provisional until additional empirical validation appears (Zhang, 2026; Joshi & Chance, 2026).

Leverage Choice, Liquidation Risk, and Lottery-like Behavior

The leverage and liquidation evidence is among the strongest direct behavioral evidence in the review because it links observed positions to economically large forced exits (Cheng et al., 2021; Alexander et al., 2023). Cheng et al. (2021) estimate that daily forced liquidations on BitMEX bitcoin perpetuals were substantial and that liquidated traders used an average leverage of 60x. The same paper estimates that margins would need to rise substantially to reduce the daily margin-call probability to 1%, indicating that prevailing leverage choices exposed traders to severe liquidation risk (Cheng et al., 2021).

Alexander et al. (2023) show that hedgers who account for automatic liquidation should use much lower leverage than traders often choose in cryptocurrency futures markets. Their model demonstrates that effective hedging requires a trade-off between hedge precision and liquidation probability (Alexander et al., 2023). Observed high leverage, therefore, fits speculative or lottery-like motives more closely than disciplined hedging when researchers

cannot identify a hedging need or collateral-management rationale (Cheng et al., 2021; Alexander et al., 2023).

Chen et al. (2024) extend the discussion of leverage and liquidation to decentralized venues by showing that trader behavior differs across CEX, oracle-pricing, and VAMM models. Their Electronic Markets article reports that oracle-priced DEXs exhibit directional asymmetry, with less-informed traders reacting more strongly to positive than to negative news (Chen et al., 2024). This finding is consistent with optimism bias or skewed belief updating. Still, the interpretation remains indirect because the study infers trader information and beliefs from observed exchange behavior rather than surveys or account-level belief measures (Chen et al., 2024).

Speculative Demand, Funding-rate Carry, and Arbitrage Behavior

The strongest perpetual-specific behavioral mechanism is the interaction between speculative demand and constrained arbitrage (Christin et al., 2023; Schmeling et al., 2023; He et al., 2022; Gornall et al., 2024). He et al. (2022) derive no-arbitrage prices and bounds for perpetual futures and show that crypto deviations from those benchmarks exceed comparable deviations in traditional currency markets. Their evidence makes the remaining spread behaviorally meaningful because the benchmark clarifies what mispricing would look like absent demand pressure or binding trading frictions (He et al., 2022).

Christin et al. (2023) formalize the crypto carry trade as a long-spot and short-perpetual strategy that harvests funding premia when long-side leverage demand pushes perpetual prices above spot. Schmeling et al. (2023) provide related evidence on crypto carry and market segmentation in a BIS working paper. Together, these papers support the interpretation that funding-rate carry can proxy for speculative demand and disagreement, although funding premia

can also reflect segmentation, collateral frictions, and arbitrage costs (Christin et al., 2023; Schmeling et al., 2023).

Gornall et al. (2024) reinforce this account by modeling constrained arbitrage capital and volatile speculative demand as drivers of futures–spot dislocations. Ackerer et al. (2025) sharpen the contract-design benchmark by showing how funding design affects price anchoring in linear, inverse, and quanto perpetuals. The combined implication is that carry returns and basis deviations become behaviorally interpretable only after the pricing benchmark and contract design are specified (Ackerer et al., 2025; Gornall et al., 2024).

Informed Trading, Market Quality, and Microstructure Behavior

The informed-trading and market-quality theme provides the strongest causal evidence in the direct perpetual literature (Ruan & Streltsov, 2022). Ruan and Streltsov (2022) exploit funding-cycle timing, Huobi’s termination of perpetual trading, and staggered contract introductions to show that perpetual contracts raise spot-market trading volume but widen quoted spreads. They resolve the apparent liquidity puzzle by documenting increased informed trading, especially around funding-settlement windows and larger funding-fee magnitudes (Ruan & Streltsov, 2022).

Alexander et al. (2024) study joint efficiency across bitcoin and ether options and perpetual futures, identifying economically meaningful arbitrage opportunities. Their evidence suggests that derivative-market inefficiency can persist when professional, informed traders interact with less-informed retail flow and transaction costs (Alexander et al., 2024). The paper does not directly measure psychology, but it links arbitrage profitability to trader composition and information asymmetry (Alexander et al., 2024).

Giagkiozis and Said (2024) show that some large derivatives exchanges report implausible open-interest dynamics or delayed liquidation messages in bitcoin perpetual swaps. This result matters for behavioral interpretation because open interest often serves as a proxy for crowding, leverage, sentiment, and liquidity (Giagkiozis & Said, 2024). If the proxy contains systematic reporting noise, studies that infer sentiment or crowding from open interest need stronger data validation and cross-venue robustness checks (Giagkiozis & Said, 2024).

Trader Heterogeneity and Learning

Trader heterogeneity is conceptually central but empirically underidentified in perpetual-futures research (Ferko et al., 2023; Alexander et al., 2024; Chen et al., 2024). The literature distinguishes among retail traders, institutional traders, informed traders, uninformed traders, arbitrageurs, directional traders, liquidity providers, and market makers, but direct perpetual datasets rarely record participant identities (Alexander et al., 2024; Chen et al., 2024). This gap limits psychological inference because the same market-level pattern can arise from different participant types (Ferko et al., 2023; Chen et al., 2024).

Alexander et al. (2024) infer trader-composition effects from arbitrage opportunities in options–perpetual markets rather than from named trader accounts. Chen et al. (2024) provide stronger evidence of design-based heterogeneity by separating traders from liquidity providers in DEX architectures and comparing CEX, oracle, and VAMM structures. Ferko et al. (2023) use regulatory data on CME bitcoin futures to show that bitcoin futures attract concentrated and diversified trader types. Still, that evidence is adjacent, as CME-dated futures differ from offshore perpetual contracts (Ferko et al., 2023).

The adjacent learning evidence suggests that cryptocurrency traders do not always update efficiently in ambiguous environments (Gemayel & Preda, 2021). Kogan et al. (2023) show that

the same retail investors can behave differently in crypto than in stocks or gold, which supports asset-class-specific behavioral interpretation. The next empirical step is to establish a direct link between these broader trader-level findings and actual perpetual-contract accounts on large CEX and DEX venues (Gemayel & Preda, 2021; Kogan et al., 2023; Chen et al., 2024).

Exchange Design and Behavioral Responses

Exchange design changes how trader behavior enters prices, liquidity, and liquidation risk (Chen et al., 2024; Ackerer et al., 2025). Chen et al. (2024) classify perpetual venues into centralized, hybrid, oracle-pricing, and VAMM-based models and show that the same trading variables can have different implications across architectures. In VAMM-based DEXs, long and short open interest can affect volatility asymmetrically because price formation depends on virtual-pool mechanics (Chen et al., 2024).

Contract design also matters because inverse, linear, and quanto perpetuals embed different collateral, payoff, and liquidation mechanics (Ackerer et al., 2025; Atzberger et al., 2024). Inverse contracts can suit hedging motives by stabilizing fiat-denominated exposure for coin-margined short positions, whereas linear contracts can be more intuitive for directional speculation (Alexander et al., 2023; Ackerer et al., 2025). The behavioral evidence on quanto perpetuals remains limited because most studies discuss their mechanics in terms of pricing, hedging, or strategy design rather than direct trader-level behavior (Atzberger et al., 2024; Ackerer et al., 2025).

CEX–DEX comparisons require matched assets, horizons, and regimes because oracle pricing, order books, VAMMs, on-chain settlement, and custody rules change trader roles and price-formation mechanisms (Chen et al., 2024). A behavioral conclusion drawn from a CEX order book may not generalize to an oracle-priced DEX where traders act more as price takers

(Chen et al., 2024). A behavioral conclusion drawn from VAMM open interest may not generalize to a centralized limit-order book because the pricing rule and liquidity provision differ by design (Chen et al., 2024).

Regime Dependence: Bull Markets, Crashes, Contagion, and Stress Events

Regime dependence appears across liquidation, funding, basis, liquidity, and stress studies (Cheng et al., 2021; He et al., 2022; Gornall et al., 2024; Joshi & Chance, 2026). Gornall et al. (2024) show that perpetuals reduce basis-trade drawdowns relative to quarterly futures during large spot-price movements and improve crisis liquidity. This finding supports the interpretation that funding design can stabilize basis risk when speculative demand and arbitrage-capital constraints become especially relevant (Gornall et al., 2024).

He et al. (2022) show that deviations from no-arbitrage benchmarks comove across cryptocurrencies and remain economically meaningful even as they shrink over time. The same paper links larger deviations to stress episodes such as the COVID crash and the FTX episode (He et al., 2022). Ruan and Streltsov (2022) also support regime dependence because their informed-trading channel strengthens when funding-fee magnitudes are larger.

Joshi and Chance (2026) add a stress-theory frontier by modeling run-like exits from perpetual markets when adverse-selection costs interact with volatility thresholds. This evidence should remain classified as frontier working-paper evidence because it is recent and less replicated than the funding-cycle and liquidation evidence (Joshi & Chance, 2026; Ruan & Streltsov, 2022; Cheng et al., 2021). Future research should therefore model bull-market optimism, crash-induced liquidations, trust shocks following exchange failures, and contagion episodes as distinct regimes rather than as a single pooled state (He et al., 2022; Gornall et al., 2024; Joshi & Chance, 2026).

Table 3*Behavioral Constructs, Measures, and Proxies*

Construct	Typical proxy	Directness	Illustrative evidence
Sentiment / directional demand	Funding rate, basis, carry, long-short imbalance	Mostly indirect	He et al. (2022); Christin et al. (2023); Ruan & Streltsov (2022)
Attention	Search/media data; trading bursts around funding windows	Mostly indirect	Dunbar & Owusu-Amoako (2023); Ruan & Streltsov (2022)
Herding/feedback	Return-chasing positions; comovement in funding and basis	Indirect	Kogan et al. (2023); He et al. (2022); Gornall et al. (2024)
Overconfidence/FOMO	Excessive leverage, one-sided long accumulation after positive news	Indirect	Odean (1998); Cheng et al. (2021); Chen et al. (2024)
Loss aversion	Liquidation-aware leverage choice; asymmetric deleveraging near thresholds	Semi-direct	Kahneman & Tversky (1979); Alexander et al. (2023)
Lottery-like motives	Extreme leverage despite high liquidation probability	Indirect but strong	Cheng et al. (2021); Gemayel & Preda (2021)
Informed trading	VPIN, wider spreads with higher volume, order-flow toxicity	Direct	Easley et al. (2012); Ruan & Streltsov (2022); Alexander et al. (2024)
Disagreement	Persistent basis deviations, carry returns, and funding dispersion	Indirect	Christin et al. (2023); He et al. (2022); Gornall et al. (2024)
Trader heterogeneity	Trader tags, COT categories, DEX roles, order-size persistence	Mixed	Ferko et al. (2023); Alexander et al. (2024); Chen et al. (2024)

Note. No proxy maps one-to-one onto a psychological construct, so construct validity depends on triangulation and institutional design.

Table 4*Methods, Identification Strategies, and Key Limitations*

Method family	Strength	Limitation	Best examples
Natural experiments	Strong causal leverage	Rare events and venue-specific external validity	Ruan & Streltsov (2022)
Staggered difference-in-differences	Useful for contract introductions	Treatment timing may be endogenous	Ruan & Streltsov (2022); Callaway & Sant'Anna (2021)
High-frequency microstructure	Strong channel identification	Weak trader identity and cross-venue comparability	Alexander et al. (2024); Giagkiozis & Said (2024)
Structural/no-arbitrage models	Clear benchmark and counterfactual	Behavior is usually inferred rather than observed	He et al. (2022); Ackerer et al. (2025); Gornall et al. (2024)
Strategy backtests	Economic significance of carry and arbitrage	Weak psychological attribution	Christin et al. (2023); Schmeling et al. (2023)
Trader-account or positioning data	Closest to actual behavior	Often adjacent rather than perpetual-specific	Kogan et al. (2023); Ferko et al. (2023); Dunbar & Owusu-Amoako (2023)
Taxonomic/Systematization of Knowledge	Clarifies design heterogeneity	Not causal by itself	Chen et al. (2024); Almeida & Gonçalves (2023)

Note. The strongest behavioral claims require either direct trader observation or designs that isolate institutional timing and market-quality channels.

Table 5*Summary of Findings by Theme and Evidential Strength*

Theme	Main result	Strongest setting	Strength
Speculative demand and basis	Funding-rate carry and basis deviations reflect speculative demand plus arbitrage limits.	CEX perpetuals	High
Leverage and liquidation	Traders often choose leverage levels that expose them to severe liquidation risk.	BitMEX and major futures venues	High
Market quality and informed trading	Perpetuals can increase volume while widening spreads through adverse selection.	Spot–perpetual interaction	High
Retail momentum and attention	Broader crypto traders exhibit momentum and attention sensitivity that may carry over to perpetuals.	Adjacent crypto and futures positioning	Medium
Trader heterogeneity	Participant type matters but remains weakly identified in direct perpetual datasets.	CEX–DEX and regulatory data	Medium-low
Exchange design	Oracle and VAMM designs generate behavioral asymmetries not captured by standard CEX models.	DEX perpetuals	Medium
Regime dependence	Stress episodes strengthen liquidation, funding, basis, and adverse-selection channels.	Crashes and funding stress	High
Classic psychological primitives	FOMO, overconfidence, and disposition effects are conceptually relevant but rarely measured directly.	Mostly indirect evidence	Low-to-medium

Note. The strength ratings reflect identification quality, data quality, construct validity, and proximity to perpetual-futures behavior.

Operationalizing Behavioral Constructs in Perpetual-Futures Research

Table 3 shows that perpetual-futures research operationalizes behavior primarily through revealed market behavior rather than direct psychological instruments (He et al., 2022; Ruan & Streltsov, 2022; Giagkiozis & Said, 2024). This measurement structure is well-suited to high-frequency exchange data but makes construct validity a central concern (Giagkiozis & Said, 2024; Almeida & Gonçalves, 2023). The most defensible studies triangulate multiple proxies or exploit institutional timing, rather than relying on a single variable in isolation (Ruan & Streltsov, 2022; He et al., 2022).

Sentiment and disagreement are best measured using funding rates, basis deviations, carry returns, and cross-asset comovement when the researcher lacks trader-belief data (Christin et al., 2023; He et al., 2022; Gornall et al., 2024). Attention is best measured by search intensity, media data, or event-time activity around scheduled funding windows, which generate

predictable monitoring incentives (Dunbar & Owusu-Amoako, 2023; Ruan & Streltsov, 2022).

Herding and momentum are best tested by examining whether lagged returns predict subsequent changes in funding, open interest, order imbalances, or trader positions (Kogan et al., 2023; Liu et al., 2022; He et al., 2022).

Overconfidence and lottery-like motives warrant particular caution, as high leverage can reflect speculation, hedging, collateral scarcity, or exchange margin conventions (Odean, 1998; Cheng et al., 2021; Alexander et al., 2023). The strongest interpretation emerges when extreme leverage coincides with frequent liquidation, weak realized performance, and persistence after losses (Cheng et al., 2021; Gemayel & Preda, 2021). Loss aversion is most credible when researchers observe asymmetric deleveraging near liquidation thresholds or model liquidation-aware hedging choices explicitly (Kahneman & Tversky, 1979; Alexander et al., 2023).

Informed trading is the most directly measurable construct because microstructure metrics can link order flow, volume, quoted spreads, and toxicity around funding windows (Easley et al., 2012; Ruan & Streltsov, 2022). Trader heterogeneity is the least resolved construct because most major perpetual venues do not disclose account-level trader identities to researchers (Alexander et al., 2024; Chen et al., 2024). Future datasets should combine account tags, order-level data, funding-window behavior, and realized gains and losses to separate retail speculation, institutional hedging, arbitrage, and market making (Ferko et al., 2023; Kogan et al., 2023; Chen et al., 2024).

Comparative Methodological Assessment

The evidence base is strongest where institutional timing or exogenous events isolate mechanisms (Ruan & Streltsov, 2022; Callaway & Sant'Anna, 2021). Funding-cycle windows,

exchange terminations, and staggered contract introductions allow researchers to distinguish volume effects from market-quality effects more clearly than cross-sectional correlations (Ruan & Streltsov, 2022). High-frequency microstructure studies also strengthen inference by connecting trading intensity, spreads, and order-flow toxicity over short horizons (Alexander et al., 2024; Giagkiozis & Said, 2024).

Structural and no-arbitrage models are strongest when they define the benchmark against which behavioral deviations are measured (He et al., 2022; Ackerer et al., 2025). These papers discipline interpretation by clarifying how perpetual prices, funding rates, and the basis should behave under frictionless or bounded-arbitrage conditions (He et al., 2022; Ackerer et al., 2025). They do not independently prove psychological mechanisms unless paired with empirical evidence on demand pressure, trader behavior, or arbitrage constraints (He et al., 2022; Gornall et al., 2024).

Data quality is a major methodological limitation because exchanges report open interest, liquidation messages, funding rates, and volume under different conventions (Giagkiozis & Said, 2024). Giagkiozis and Said (2024) show that open-interest reporting can be implausible or delayed on some venues. This evidence implies that cross-exchange behavioral inference requires direct data validation before researchers treat open interest as evidence of sentiment, crowding, or liquidity (Giagkiozis & Said, 2024).

External validity remains uneven across assets, venues, contract types, and regimes (Chen et al., 2024; Gornall et al., 2024; Joshi & Chance, 2026). Evidence from Bitcoin Perpetuals on large centralized exchanges may not generalize to smaller altcoins, quanto contracts, oracle-priced DEXs, or VAMM-based DEXs because payoff mechanics and trader composition differ across settings (Ackerer et al., 2025; Chen et al., 2024). Methodological strength,

therefore, depends on both identification design and representativeness of the market setting (Table 4).

Integrative Framework of Behavioral Channels in Perpetual Markets

The reviewed literature supports a three-layer framework that links belief formation, design transmission, and market outcomes (Figure 2). The first layer is belief formation, where traders respond to prices, attention, disagreement, and news in ways that can appear as momentum-like crypto demand or speculative derivatives positioning (Almeida & Gonçalves, 2023; Kogan et al., 2023; Dunbar & Owusu-Amoako, 2023). In perpetual markets, these beliefs become observable through funding pressure, basis deviations, open interest, and directional leverage demand (Christin et al., 2023; He et al., 2022).

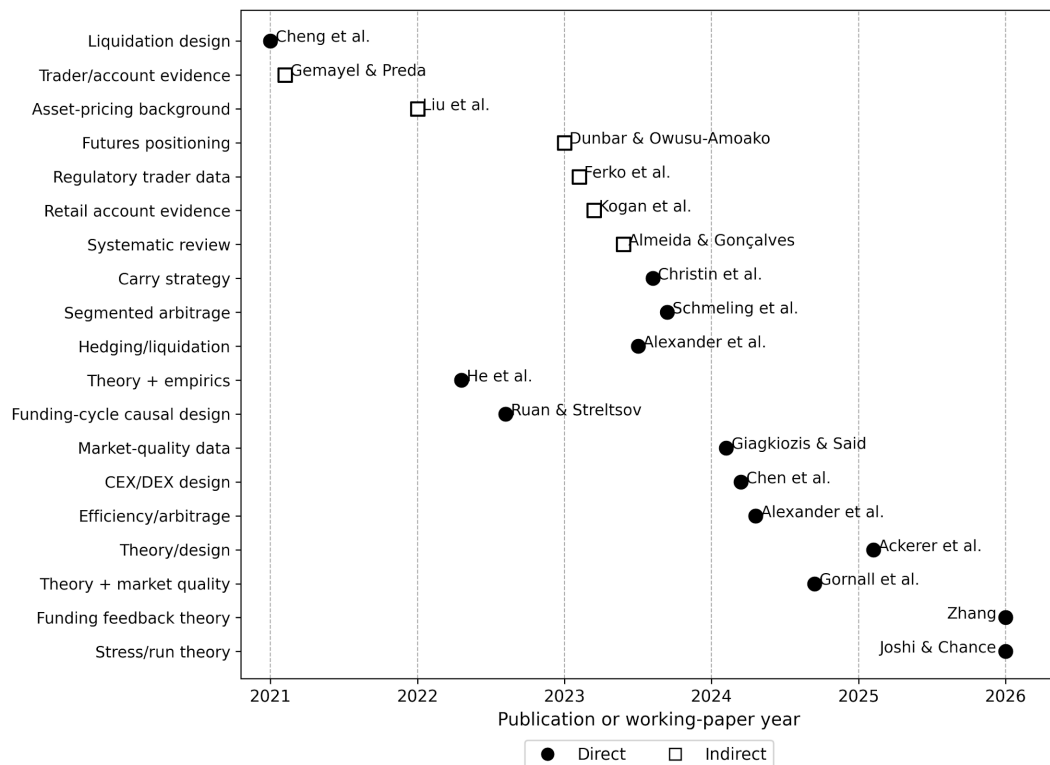
The second layer is design transmission, where funding, margin, liquidation, and venue architecture translate trader behavior into market variables (Ackerer et al., 2025; Chen et al., 2024). Funding rates periodically convert imbalances into cash transfers, while liquidation rules trigger forced exits when collateral falls below a threshold due to adverse price moves (Cheng et al., 2021; He et al., 2022). CEX, oracle pricing, and VAMM structures then determine how those pressures affect price formation, liquidity, and volatility (Chen et al., 2024).

The third layer is market outcomes, where volume, spreads, adverse selection, basis risk, carry returns, and spot spillovers reflect the interaction between beliefs and design (Ruan & Streltsov, 2022; Gornall et al., 2024). Perpetuals can increase spot-market activity while worsening adverse selection when informed trading intensifies around funding windows (Ruan & Streltsov, 2022). Perpetuals can also reduce extreme basis risk relative to quarterly futures when frequent funding payments discipline noisy speculative demand (Gornall et al., 2024).

Figure 3 places the reviewed studies in chronological and methodological context. The timeline shows that early evidence combined liquidation studies, trader-learning studies, broad crypto risk-factor work, and early perpetual-futures working papers, while later work increasingly examined CEX–DEX design, journal-published pricing models, basis risk, and stress dynamics (Cheng et al., 2021; Liu et al., 2022; He et al., 2022; Ruan & Streltsov, 2022; Chen et al., 2024; Ackerer et al., 2025; Gornall et al., 2024; Joshi & Chance, 2026). This sequence shows a field moving from adjacent behavioral foundations toward direct perpetual-market identification (Figure 3).

Figure 3

Timeline of the Reviewed Literature by Method and Directness



Note. Filled circles indicate direct perpetual-futures studies, and open squares indicate adjacent behavioral or asset-pricing studies; the figure uses marker shape rather than color to remain

interpretable in grayscale (Table 2). The year reflects the publication year for published work and the working-paper year for unpublished work in the reference list.

Research Gaps and Future Research Agenda

Trader-level identification is the central missing piece because most direct perpetual studies infer behavior from funding, basis, open interest, volume, spreads, and liquidation series rather than from trader identities or account decisions (Cheng et al., 2021; Giagkiozis & Said, 2024; He et al., 2022). Adjacent studies provide stronger trader-level evidence, but they typically examine retail crypto accounts, CME futures, or broader cryptocurrency behavior rather than perpetual accounts on major CEX and DEX venues (Ferko et al., 2023; Kogan et al., 2023; Almeida & Gonçalves, 2023). Future research should prioritize account-linked exchange data or quasi-account-level aggregates that distinguish retail traders, institutions, market makers, arbitrageurs, liquidity providers, and directional traders (Ferko et al., 2023; Chen et al., 2024).

Causal tests of leverage-rule changes are needed because leverage and liquidation are central to perpetual-market behavior, yet they have not yet been studied with sufficient variation in exogenous rules (Cheng et al., 2021; Alexander et al., 2023). The current strongest causal evidence concerns funding-cycle timing, exchange terminations, and contract introductions rather than leverage-cap interventions (Ruan & Streltsov, 2022). Future work should exploit discrete changes in leverage caps, maintenance-margin rules, funding caps, or liquidation procedures to identify how traders respond when exchanges alter the effective gambling intensity of perpetual markets (Cheng et al., 2021; Alexander et al., 2023).

CEX–DEX comparison remains underdeveloped because DEX evidence is newer and contract architectures differ sharply across oracle-pricing and VAMM designs (Chen et al., 2024). Evidence is more developed for centralized exchanges on funding, basis, arbitrage, and

market quality than for decentralized perpetual venues (Alexander et al., 2024; Ruan & Streltsov, 2022; Gornall et al., 2024). Future work should compare the same token and contract type across CEX and DEX venues with matched horizons, matched market regimes, and validated open-interest measures (Giagkiozis & Said, 2024; Chen et al., 2024).

Contract-type heterogeneity remains understudied because linear, inverse, and quanto contracts embed different collateral, settlement, and liquidation mechanics (Atzberger et al., 2024; Ackerer et al., 2025). Existing hedging and pricing studies show that these mechanics change hedging performance and price anchoring, but they do not yet identify whether traders self-select into contract types based on behavioral traits or risk preferences (Alexander et al., 2023; Ackerer et al., 2025). Future behavioral work should test whether leverage choice, liquidation frequency, funding sensitivity, and realized performance differ across linear, inverse, and quanto products (Atzberger et al., 2024; Ackerer et al., 2025).

Behavioral constructs need cleaner measurement because funding, basis, open interest, volume, spreads, and liquidations do not map uniquely onto sentiment, disagreement, loss aversion, overconfidence, or lottery demand (He et al., 2022; Ruan & Streltsov, 2022; Almeida & Gonçalves, 2023). Future studies should combine market data with direct or semi-direct measures such as search intensity around funding windows, wallet-linked repeat trading, realized gain and loss histories, leverage histories, and account-level liquidations (Dunbar & Owusu-Amoako, 2023; Kogan et al., 2023; Cheng et al., 2021). These measures would reduce overinterpretation of market-level proxies and improve construct validity (Almeida & Gonçalves, 2023; Giagkiozis & Said, 2024).

Regime dependence should be modeled explicitly because stress sensitivity appears in liquidation, basis, liquidity, and informed-trading studies (Cheng et al., 2021; He et al., 2022;

Ruan & Streltsov, 2022; Gornall et al., 2024). The literature should distinguish among bull-market optimism, crash-induced liquidations, exchange-failure trust shocks, contagion episodes, and stablecoin-related funding stress, as behavioral responses may differ across regimes (He et al., 2022; Gornall et al., 2024; Joshi & Chance, 2026). A regime-aware approach would help distinguish ordinary speculative demand from stress-driven arbitrage-capital withdrawal (Gornall et al., 2024; Joshi & Chance, 2026).

Limitations

The review is constrained rather than fully database-native because it did not query Web of Science and Scopus directly (Page et al., 2021; Table 1). This limitation can reduce source-platform completeness even though the review used publisher pages, SSRN, NBER, BIS, AEA, or author pages, and citation trails to compensate (Appendix A). The synthesis should therefore be read as transparent and replicable within the accessible search environment rather than exhaustive across all bibliographic databases (Page et al., 2021).

The evidence hierarchy also limits causal interpretation because several direct perpetual studies rely on structural models, strategy backtests, or taxonomies rather than quasi-experimental designs (He et al., 2022; Akerer et al., 2025; Chen et al., 2024). The strongest causal claims come from funding-cycle, exchange-termination, and contract-introduction designs, but those settings may not generalize to all assets, venues, and regimes (Ruan & Streltsov, 2022; Gornall et al., 2024). The review therefore distinguishes direct causal evidence, direct descriptive evidence, direct theory, and adjacent behavioral evidence throughout the synthesis (Table 2; Table 4).

Working-paper evidence remains important but provisional because several frontier studies on funding design, basis risk, and stress dynamics have not yet completed peer review

(Gornall et al., 2024; Zhang, 2026; Joshi & Chance, 2026). The review uses those papers to identify mechanisms and research gaps. Still, it does not treat their claims as well established as those in peer-reviewed studies on liquidation, hedging, exchange design, and market quality (Cheng et al., 2021; Alexander et al., 2023; Chen et al., 2024; Ruan & Streltsov, 2022). This distinction protects the synthesis from overstating emerging evidence (Table 2).

Conclusion

This constrained systematic review shows that behavioral finance matters in cryptocurrency perpetual futures, but it usually enters the literature through market structure rather than direct psychological measurement (Almeida & Gonçalves, 2023; He et al., 2022; Ruan & Streltsov, 2022). The strongest direct evidence concerns speculative demand, leverage choice, liquidation pressure, funding-rate carry, arbitrage constraints, informed trading, and exchange design (Cheng et al., 2021; Christin et al., 2023; Chen et al., 2024; Gornall et al., 2024). Classic psychological constructs such as FOMO, overconfidence, disposition effects, and learning remain more weakly measured because most direct perpetual datasets do not observe beliefs, account identities, or realized trader histories (Chen et al., 2024; Gemayel & Preda, 2021; Kogan et al., 2023).

The answer to RQ1 is that the direct perpetual literature identifies behavior mainly through funding, basis, liquidation, leverage, carry, spreads, VPIN, and open interest rather than through surveys or psychometrics (Cheng et al., 2021; He et al., 2022; Ruan & Streltsov, 2022; Giagkiozis & Said, 2024). The answer to RQ2 is that funding rules, margin systems, liquidation procedures, and CEX–DEX architecture transmit behavior into observable market outcomes (Ackerer et al., 2025; Chen et al., 2024). The answer to RQ3 is that funding-cycle causal designs and high-frequency microstructure studies provide the strongest current evidence, whereas

structural and adjacent behavioral studies offer useful but less direct interpretations (Alexander et al., 2024; He et al., 2022; Ruan & Streltsov, 2022).

The answer to RQ4 is that the field needs trader-level data, leverage-rule experiments, matched CEX–DEX comparisons, contract-type heterogeneity tests, and regime-aware models of stress and arbitrage-capital withdrawal (Cheng et al., 2021; Ferko et al., 2023; Chen et al., 2024; Gornall et al., 2024; Joshi & Chance, 2026). The future research agenda should identify who trades perpetuals, which contract designs they choose, how much leverage they use, and how their positions move through funding, liquidations, and market-maker responses into basis dynamics and market quality (Alexander et al., 2024; Chen et al., 2024; Ruan & Streltsov, 2022). This agenda connects behavioral finance, derivative design, and crypto market microstructure at the point where beliefs become funding flows, margin calls, and price discovery (Ackerer et al., 2025; He et al., 2022; Ruan & Streltsov, 2022).

Glossary

Behavioral finance terms

Adverse selection. Adverse selection is the risk that liquidity suppliers trade against better-informed traders; in perpetual-linked markets, it is inferred from spreads widening despite higher volume during information- or funding-intensive periods (Glosten & Milgrom, 1985; Ruan & Streltsov, 2022).

Anchoring. Anchoring occurs when decision makers rely too heavily on an initial reference value and adjust insufficiently when new information arrives (Tversky & Kahneman, 1974).

Attention. Investor attention is the scarce cognitive focus that investors allocate to assets or information; empirical finance often measures it through news, search volume, trading activity, or salient price moves (Barber & Odean, 2008; Da et al., 2011).

Behavioral finance. Behavioral finance examines how psychological, cognitive, emotional, and social factors shape financial choices and market prices when investors deviate from strict rationality (Barber & Odean, 2000; Kahneman & Tversky, 1979).

Disagreement. Disagreement denotes heterogeneity in beliefs about asset value or future returns; when short-sale, leverage, or arbitrage constraints bind, disagreement can affect prices, trading volume, and derivative premia (Christin et al., 2023; Miller, 1977).

Disposition effect. The disposition effect is the tendency to sell winning positions too early and hold losing positions too long relative to a rational tax or performance benchmark (Odean, 1998a; Shefrin & Statman, 1985).

Fear of missing out (FOMO). FOMO is anxiety that others are obtaining rewarding experiences or gains while one is absent; in crypto-investment settings, it can mediate the effects of

herding, loss aversion, and overconfidence on investment decisions (Kaur et al., 2024; Klepacki, 2025; Przybylski et al., 2013).

Feedback trading. Feedback trading is trading that responds to prior price movements and can reinforce trends when past returns change demand, beliefs, or perceived adoption probabilities (De Long et al., 1990; Hong & Stein, 1999).

Gambling or lottery-like motives. Gambling or lottery-like motives describe a preference for assets or strategies with small probabilities of very large gains despite unfavorable risk-adjusted expectations; in perpetual markets, this interpretation is most plausible when extreme leverage coincides with frequent liquidation risk (Barberis & Huang, 2008; Cheng et al., 2021; Kumar, 2009).

Herdning. Herding occurs when investors imitate others' trades or beliefs rather than relying solely on private information, which can lead to correlated actions and amplify price movements (Almeida & Gonçalves, 2023; Bikhchandani et al., 1992).

Heuristics. Heuristics are judgment shortcuts that simplify decisions under uncertainty but can create systematic biases, including anchoring, availability, and representativeness (Tversky & Kahneman, 1974).

Informed trading. Informed trading occurs when some traders use superior or timelier information; researchers infer it from adverse-selection measures, order-flow toxicity, price discovery, and spread responses (Easley et al., 2012; Kyle, 1985; Ruan & Streltsov, 2022).

Learning. Learning refers to updating beliefs, expectations, or trading behavior based on prior outcomes; in cryptocurrency markets, it is difficult to identify without trader-level

histories that link experience, performance, and subsequent choices (Gemayel & Preda, 2021; Kogan et al., 2023).

Loss aversion. Loss aversion is the prospect-theory tendency to weigh losses more heavily than equivalent gains relative to a reference point (Kahneman & Tversky, 1979; Kaur et al., 2024).

Momentum trading. Momentum trading is a strategy or behavior in which investors buy, hold, or increase exposure after positive returns and sell, avoid, or reduce exposure after negative returns (Jegadeesh & Titman, 1993; Kogan et al., 2023).

Noise trading. Noise trading refers to trading based on sentiment, mistaken beliefs, liquidity needs, or non-fundamental signals; such demand can affect prices when arbitrage is risky or capital constrained (De Long et al., 1990; Shleifer & Vishny, 1997).

Overconfidence. Overconfidence is excessive confidence in one's information, judgment, or trading skill; models and household-account evidence link it to excessive turnover and lower net returns (Barber & Odean, 2000; Odean, 1998b).

Prospect theory. Prospect theory is a descriptive model of choice under risk in which people evaluate gains and losses relative to a reference point, exhibit loss aversion, and apply nonlinear decision weights to probabilities (Kahneman & Tversky, 1979).

Representativeness. Representativeness is a heuristic in which people judge probabilities by how similar a case is to a category, pattern, or recent trend; in trading contexts, it can support extrapolation when recent returns are interpreted as evidence of a persistent state (Kogan et al., 2023; Tversky & Kahneman, 1974).

Sentiment. Investor sentiment refers to beliefs, optimism, pessimism, or demand pressure that is not fully justified by fundamentals; in cryptocurrency research, it is often inferred from

attention, trading behavior, social signals, funding, or positioning (Almeida & Gonçalves, 2023; Baker & Wurgler, 2006).

Speculative demand. Speculative demand is demand for exposure motivated by expected price changes rather than by hedging or use value; in perpetual markets,, it can appear as basis, funding premia, leverage, and open interest (Christin et al., 2023; Gornall et al., 2024).

Trader heterogeneity. Trader heterogeneity refers to differences in trader types, information, objectives, sophistication, and constraints, such as retail versus professional traders, informed versus uninformed traders, arbitrageurs versus directional traders, and traders versus liquidity providers (Alexander et al., 2024; Chen et al., 2024; Kogan et al., 2023).

Perpetual futures and market-structure terms

Basis. The basis is the difference between a derivative price and the spot price of the underlying asset; in perpetual markets, funding mechanisms and arbitrage strategies are designed to reduce this gap, although speculative demand and arbitrage limits can keep it economically meaningful (He et al., 2022; Gornall et al., 2024).

Centralized exchange (CEX). A centralized exchange is a venue in which a central operator manages custody, order matching, risk controls, and liquidation procedures, which makes order-book design and margin rules central to perpetual-market behavior (Chen et al., 2024; Giagkiozis & Said, 2024).

Decentralized exchange (DEX). A decentralized exchange executes trading through smart contracts. It can use oracle pricing, VAMM pricing, or hybrid mechanisms, which change trader roles, liquidity provision, and price discovery relative to CEX order books (Chen et al., 2024).

Funding rate. The funding rate is the periodic transfer between long and short perpetual positions that incentivizes convergence between the perpetual price and the spot price (Ackerer et al., 2025; Christin et al., 2023; He et al., 2022).

Inverse perpetual. An inverse perpetual is typically coin-margined, so its collateral, payoff, and liquidation dynamics depend on the underlying asset price and can differ from those of linear stablecoin-margined contracts (Ackerer et al., 2025; Atzberger et al., 2024).

Linear perpetual. A linear perpetual is typically quoted and margined in a stablecoin or fiat-like unit, which can make collateral accounting more intuitive but does not remove liquidation risk under high leverage (Ackerer et al., 2025; Alexander et al., 2023).

Liquidation. Liquidation is the forced closure of a position when collateral falls below maintenance margin requirements, and liquidation risk is central to both hedging and speculative leverage decisions in bitcoin futures and perpetuals (Cheng et al., 2021; Alexander et al., 2023).

Maintenance margin. Maintenance margin is the minimum collateral level required to keep a leveraged position open, and falling below it makes a position eligible for liquidation under exchange rules (Cheng et al., 2021; Chen et al., 2024).

Open interest. Open interest is the number of outstanding derivative contracts at a point in time, and perpetual futures researchers use it as a proxy for activity, crowding, liquidity, and leverage, even though exchange reporting can be noisy (Giagkiozis & Said, 2024).

Oracle-pricing model. An oracle-pricing DEX uses an external price rule rather than only an internal order book, which can make traders behave more like price takers and can generate directional asymmetries after news (Chen et al., 2024).

Perpetual future / perpetual swap. Perpetuals are derivative contracts on an underlying asset that (a) have no contractual maturity, (b) are marked to market through an exchange-defined mark price, (c) require margin collateral and allow leverage, and (d) include periodic funding payments between longs and shorts that depend on the perpetual–spot spread and often include an interest-rate component. (Neubert et al., 2026a, pp. 3–4).

Quanto perpetual. A quanto perpetual uses an underlying, quote currency, and settlement or collateral currency that are not all identical, so payoff and liquidation dynamics depend on more than one price process (Atzberger et al., 2024; Ackerer et al., 2025).

VAMM. A virtual automated market maker uses an algorithmic pricing rule rather than a conventional order book, and VAMM-based perpetual DEXs can create asymmetric relationships between long and short open interest and volatility (Chen et al., 2024).

VPIN. VPIN is a market microstructure measure of order-flow toxicity that researchers use to infer informed trading and adverse selection risk in high-frequency markets (Easley et al., 2012; Ruan & Streltsov, 2022).

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Appendix A. Search Log and Source Verification

The appendix documents the sources used to verify retained studies and separates source verification from the substantive literature synthesis (Page et al., 2021).

Table A1

Source Verification Log

Study	Verification source	Revision action
Ackerer et al. (2025)	Publisher DOI / Mathematical Finance	Verified as a peer-reviewed journal article.
Alexander et al. (2023)	Publisher DOI / European Journal of Operational Research	Verified as a peer-reviewed journal article.
Alexander et al. (2024)	Publisher DOI / Journal of Financial Markets	Verified as a peer-reviewed journal article.
Almeida & Gonçalves (2023)	Publisher DOI / Journal of Behavioral and Experimental Finance	Verified as a peer-reviewed journal article.
Atzberger et al. (2024)	IEEE DOI	Verified as a conference proceeding.
Cheng et al. (2021)	Taylor & Francis DOI / Applied Economics	Verified as a peer-reviewed journal article.
Chen et al. (2024)	Springer DOI / Electronic Markets	Updated from SSRN working paper to peer-reviewed journal article.
Christin et al. (2023)	Author-hosted working paper	Verified as working paper; not treated as peer-reviewed evidence.
Dunbar & Owusu-Amoako (2023)	Publisher DOI / Journal of Behavioral and Experimental Finance	Verified as a peer-reviewed journal article.
Ferko et al. (2023)	Publisher DOI / Global Finance Journal	Verified as a peer-reviewed journal article.

Study	Verification source	Revision action
Giagkiozis & Said (2024)	Ledger DOI	Verified as a peer-reviewed journal article.
Gornall et al. (2024)	SSRN DOI	Verified as working paper; date follows SSRN date written, with last revision in 2025.
He et al. (2022)	SSRN DOI and arXiv record	Verified as working paper; SSRN lists first posting in 2022 and last revision in 2023.
Joshi & Chance (2026)	SSRN DOI	Verified as working paper; treated as frontier evidence.
Kogan et al. (2023)	NBER DOI	Verified as an NBER working paper.
Ruan & Streltsov (2022)	SSRN DOI	Verified as working paper; SSRN lists first posting in 2022 and last revision in 2025.
Schmeling et al. (2023)	BIS Working Paper page	Verified as BIS Working Paper No. 1087.
Zhang (2026)	SSRN DOI	Verified as working paper; treated as frontier theory evidence.

Note. The log records verification status rather than evidence quality; evidence quality is summarized in Tables 2 and 4.

Declarations

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Conflict of (Competing) Interest

The authors declare that they have no competing financial or non-financial interests related to this study.

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Data Availability and Supplementary Material

All data generated and analyzed during this study and supplementary material are available upon reasonable request.

Prior Publication

The authors confirm that this research has not been published previously and is not under consideration for publication elsewhere.

Ethics Statement

This study complies with the ethical guidelines of the European Code of Conduct for Research Integrity and adheres to GDPR requirements for data protection.

Responsible AI Ethics Statement

This study used artificial intelligence tools to support tasks such as identifying relevant literature, analyzing datasets, translating, and editing textual content. These tools were employed to enhance efficiency, and their outputs were critically reviewed to ensure alignment with research objectives. The authors made all final decisions and retain full responsibility for the integrity, rigor, and conclusions of this research.

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